Problem 1

Use Langrange's theorem in the multiplicative group $(\mathbb{Z}_p)^{\times}$ to prove fermat's little theorem: if p is a prime then $a^p \equiv a \bmod p$ for all $a \in \mathbb{Z}$

Solution:

Consider that $a^p = a^{p-1} a \bmod p$

Then if we show that $a^{p-1} = 1 \bmod p$ for any p then the claim follows

To see why this is true, note that for an element $a \in \mathbb{Z}_p^{\times}$, and by a corollary to langrange's theorem, $|a| \mid |\mathbb{Z}_p^{\times}| = p - 1 \Rightarrow |a| \ k = p - 1$ for some k

Then taking raising both sides by k gives

$$a^{(p-1)/k} = 1 \Rightarrow a^{p-1} = 1 \bmod p$$

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Problem 2

Use Langrange's theorem in multiplicative group $(\mathbb{Z}_n)^{\times}$ to prove Euler's theorem: $a^{\varphi(n)} \equiv 1 \bmod n$ for every a relatively prime to n, where φ denotes Euler's φ function

Solution:

Note that the order of $\mathbb{Z}_n$ is $\varphi(n)$

Then by a corollary to lagrange's theorem for any $a \in \mathbb{Z}_n^{\times}$ we have $|a| \mid |\mathbb{Z}_n| \Rightarrow |a|k = \varphi(n) \Rightarrow a^{\varphi(n)/k} = 1 \Rightarrow a^{\varphi(n)} = 1 \bmod n$ after raising both sides by k

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Problem 3

1. prove that if H, K are normal subgroups of a group G then their intersection $H \cap K$ is also a normal subgroup of G
2. Prove that the intersection of an arbitary non-empty collection of normal subgroups of a group is a normal subgroup (do not assume that the collection is countable)

Solution:

1.

Note that $H \cap K \leq G$ by an earlier homework, so we need to show that $H \cap K$ is a normal subgroup of G

In order to show that $g(H \cap K)g^{-1} = H \cap K$ it is enough (from a proposition in class) that $g(H \cap K)g^{-1} \subseteq H \cap K$

For some $g \in G$ any any $x \in H \cap K$ we have that $gxg^{-1} \in g(H \cap K)g^{-1}$

Since $H \trianglelefteq G$ then $gxg^{-1} \in H$ and since $K \trianglelefteq G$ then $gxg^{-1} \in K$

Then we have $gxg^{-1} \in H \cap K$ and the claim follows

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2.

From a previous homework we showed that the intersection of an arbitrary non-empty collection of subgroups of a group G is again a subgroup of G

Then we need to show this intersection is normal i.e. that if $H = \bigcap_{a \in \mathrm{A}} H_\alpha$ where each H_α s.t. $a \in \mathrm{A}$ is a normal subgroup of G then H is a normal subgroup of G (i)

It is enough to show that $g_1(\bigcap_{a \in \mathrm{A}} H_\alpha)g_1^{-1} \subseteq \bigcap_{a \in \mathrm{A}} H_\alpha$

let $g_1 \in G$ and any $x \in \bigcap_{a \in \mathrm{A}} H_\alpha$ then $g_1 x g_1^{-1} \in g_1(\bigcap_{a \in \mathrm{A}} H_\alpha)g_1^{-1}$

Note that if $x \in \bigcap_{a \in \mathrm{A}} H_\alpha$ then $x \in H_\alpha$ for any α and that x has the form $x = g_2 y g_2^{-1}$ for $y \in H_\alpha$ any $g_2 \in G$

Then $g_1 x g_1^{-1} = g_1(g_2 y g_2^{-1})g_1^{-1} = (g_1 g_2) y (g_1 g_2)^{-1} \in H_\alpha$ for any $\alpha \therefore g_1 x g_1^{-1} \in \bigcap_{a \in \mathrm{A}} H_\alpha$ (the normal subgroups definition holds for any $g \in G$ (note that $g_1 g_2 \in G$))

The claim follows

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(i) A contains the indices which "label" each H_α

Problem 4

Prove that if $N \trianglelefteq G$ and H is any subgroup of G then $N \cap H \trianglelefteq H$

Solution:

It is enough to show that for any h that $h(N \cap H)h^{-1} \subseteq N \cap H$

let $h \in H$ and $x \in N \cap H$ then $hxh^{-1} \in h(N \cap H)h^{-1}$

Note that $hxh^{-1} \in N$ since $N \trianglelefteq G, h \in G, h^{-1} \in G, x \in N$ since $H \subseteq G$

Note that $hxh^{-1} \in H$ since $h \in H, h^{-1} \in H, x \in H$ and that H must be closed

Then $hxh^{-1} \in N \cap H$ and the claim follows

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Problem 5

Let N be a finite subgroup of a group G and assume that $N = \langle S \rangle$ for some subset S of G. Prove that an element $g \in G$ normalises N iff $gSg^{-1} \subseteq N$

Solution:

This is equivalent to showing that $gNg^{-1} = N \Leftrightarrow gSg^{-1} \subseteq N$

$(\Rightarrow)$ Assume that $gNg^{-1} = N$

Then $gng^{-1} \in N$ for any $n \in N$

Consider $gsg^{-1} \in gSg^{-1}$ for any $s \in S$

Then $gsg^{-1} \in N$ since $s \in S \Rightarrow s \in N \because N = \langle S \rangle$

$(\Leftarrow)$ Want to show that $gSg^{-1} \subseteq N \Rightarrow gNg^{-1} = N$

Firstly, note that for any $s \in S$ we have $(gsg^{-1})(gs^{-1}g^{-1}) = 1$. Since $gsg^{-1} \in N$ then the inverse should also be in N i.e. $gs^{-1}g^{-1} \in N$ (0)

Next, denote an element in n to be $n = s_1^{\varepsilon_1} s_2^{\varepsilon_2} ... s_m^{\varepsilon_m}$ where $\varepsilon = \pm 1$ by the definition of groups generated by a subset

Now, consider $gng^{-1} \in gNg^{-1}$ for some $n \in N$

By a nice trick,

$$gng^{-1} = gs_1^{\varepsilon_1} s_2^{\varepsilon_2} ... s_m^{\varepsilon_m} g^{-1} = (gs_1^{\varepsilon_1} g^{-1})(gs_2^{\varepsilon_2} g^{-1}) ... (gs_m^{\varepsilon_m} g^{-1}) \in N$$

since each term is in N by (0) and that N is a closed (1)

Then, consider $n \in N$ then

$$n = s_1^{\varepsilon_1} s_2^{\varepsilon_2} ... s_m^{\varepsilon_m} = g(g^{-1} s_1^{\varepsilon_1} g g^{-1} s_2^{\varepsilon_2} g ... g^{-1} s_m^{\varepsilon_m} g) g^{-1} = g(g^{-1} n g) g^{-1}$$

And by the previous result (1) we know that $g^{-1}ng \in N$ therefore n can be written as $gng^{-1} \Rightarrow n \in gNg^{-1}$ (2)

The claim follows by (1) and (2)

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Problem 6

Prove that the group of inner automorphisms $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$

Solution:

Note that $\mathrm{Inn}(G) = \{\phi_g : \phi_g(x) = gxg^{-1} \forall x \in G\}$ where $\phi_g : G \to G$

This is a group with identity ϕ_1 and inverse $\phi_{g^{-1}}$ and a subset of $\mathrm{Aut}(G)$ (we showed ϕ_g is an automorphism in class)

We would like to show that for all $\varphi \in \mathrm{Aut}(G)$ that $\varphi \phi_g \varphi^{-1} \in \mathrm{Inn}(G)$. Note that

$$(\varphi \phi_g \varphi^{-1})(x) = \varphi(\phi_g(\varphi^{-1}(x))) = \varphi(g \varphi^{-1}(x) g^{-1}) = \varphi(g) x \varphi(g^{-1}) = \varphi(g) x (\varphi(g))^{-1} = \phi_{\varphi(g)} \in \mathrm{Inn}\ (G)$$

since φ is a homomorphism and $\varphi(x^{-1}) = (\varphi(x))^{-1}$ for any $x \in G$

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Problem 7

Let G be a gorup such that $|G| = pq$ where p, q are prime. Prove that every proper subgroup of G is cyclic

Solution:

The divisors of pq are $1, p, q, pq$

Then any proper subgroup of G must have order $1, p, q$ by lagrange's theorem

We know that $\{1\}$ is cyclic, and that groups of prime order are also cyclic (from homework 5 problem 9)

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Problem 8

Let G be a group such that $|G| = 8$. Show that G must have an element of order 2

Solution:

By lagrange's theorem, any $g \in G$ must have order $1, 2, 4, 8$

Suppose that there does not exist $g \in G$ such that $|g| = 2$

Since $|G| = 8$ then there exist $g \in G$ such that $g \neq 1$

Suppose $|g| = 8$ then $G = \langle g \rangle$ and $g^4 \in G$ has order 2

Suppose $|g| = 4$ then $\{e, g, g^2, g^3\} \leq G$ and $\left(g^2\right)^2 = 1$ and by assumption $g^2 \neq 1 \Rightarrow |g^2| = 2$

These are both contradictions, so G must have an element of order 2

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Problem 9

Let G be an abelian group with odd order. Show that the product of all the elements of G is the identity

Solution:

By Lagrange's theorem, there cannot exist an element of order 2

Then every element $g \in G$ has an inverse which is not itself i.e. $g^{-1} \neq g$ (except $1 \in G$)

Then $\prod_{g \in G} g = 1$ due to G being abelian (order of multiplication does not matter).

In other words, if we build this product on term at a time, we can pair each term with its inverse and also multiply by the identity for a total of $2k + 1$ terms for some k

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Problem 10

(Will not be graded: classification of groups of order $2p$. Let G be a group of order $2p$ where p is a prime greater than 2. Show that G is isomorphic to either $\mathbb{Z}_{2p}$ or D_{2p}.)